

Behavioral Drivers of Online Purchase Conversion in Session-Level E-Commerce Data

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Exploratory Data Analysis and Visualization

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Table of Contents

Chapter 1: Introduction

- 1.1 Background and Motivation
- 1.2 Dataset and Analytical Context
- 1.3 Research Questions
- 1.4 Scope and Organization

Chapter 2: Data Description, Audit, and Cleaning

- 2.1 Data Source
- 2.2 Variable Overview and Data Dictionary
- 2.3 Key Distributional Properties and Business Implications
- 2.4 Data Quality Audit and Cleaning Pipeline
- 2.5 Dataset Limitations

Chapter 3: Univariate Exploration

- 3.1 Outcome and Audience Composition
- 3.2 Product Engagement Metrics
- 3.3 Session Context Indicators

Chapter 4: Bivariate and Multivariate Analysis

- 4.1 Product Engagement and Conversion
- 4.2 Visitor Type, Day Type, and Bounce Behavior
- 4.3 Traffic Source, Geography, and Purchase Outcomes
- 4.4 Multivariate Structure: Scatter Matrix
- 4.5 Summary

Chapter 5: Feature Engineering

5.1 Motivation

5.2 Log Transformations and Ratio Features

5.3 Temporal and Binned Features

5.4 Feature Value Assessment

5.5 Updated Feature Vector

Chapter 6: Descriptive Modeling

6.1 Logistic Regression: What Predicts Purchase?

6.2 Linear Regression: What Drives Browsing Depth?

6.3 Two-Way ANOVA: Visitor Type and Day Type on PageValues

6.4 K-Means Clustering: Behavioral Segmentation

6.5 Modeling Summary

Chapter 7: Visualization Design and Data Storytelling

7.1 Color Contrast

7.2 Axis Scale Distortion

7.3 Auto-Generated Labels and Text Overlap

7.4 Summary of Design Decisions

Chapter 8: Key Findings, Limitations, and Future Work

8.1 Key Findings

8.2 Limitations

8.3 Future Work

References

Chapter 1: Introduction

1.1 Background and Motivation

Online purchase conversion rates industry-wide sit below 5%. For every hundred sessions a retailer attracts, fewer than five end in a transaction. The gap between traffic and revenue is persistent and poorly understood at the session level.

Aggregate sales reports provide information about overall sales performance, but they offer limited insight into the behavior of individual visitors before a purchase occurs. Session-level data allows researchers to examine browsing patterns, page engagement, and exit behavior at a more detailed level. Understanding these behavioral signals may help explain why some sessions end in purchases while others do not.

One variable in this dataset makes the problem concrete. PageValues has a mean of 5.89 and a median of zero. Seventy-five percent of all sessions never reach a page with any assigned commercial value. Most visitors exit the funnel before encountering the content most likely to produce a sale. Understanding where that exit happens, and among which visitor types, is the motivating problem this paper addresses.

1.2 Dataset and Analytical Context

The dataset contains 12,330 web sessions from an operational e-commerce site, deposited in the UCI Machine Learning Repository. Each record captures page counts across three content categories, time-on-page durations, Google Analytics engagement metrics (BounceRates, ExitRates, PageValues), a proximity-to-holiday indicator, and categorical context variables covering month, visitor type, traffic source, region, and day type. The binary outcome variable Revenue marks whether the session ended in a purchase. Only 15.5% did.

That figure defines the analytical challenge. A model predicting "no purchase" for every session achieves 84.5% accuracy and provides zero insight. Meaningful analysis requires

identifying which behavioral patterns, observable before the purchase decision, distinguish converting sessions from non-converting ones.

Six continuous variables exhibit severe right skew. ProductRelated, ProductRelated_Duration, and PageValues all have medians far below their means. Log($x + 1$) transformations are applied throughout. Their effect on predictive content is evaluated explicitly in Chapter 5 rather than assumed.

1.3 Research Questions

Five questions structure the inquiry.

RQ1 asks whether product page engagement predicts conversion and how strongly PageValues mediates that relationship.

RQ2 asks whether returning visitors convert at higher rates than new visitors, and whether weekend sessions or holiday proximity modulate that difference. Returning visitors account for 85.6% of all sessions — whether their numerical dominance reflects a genuine conversion advantage is tested directly.

RQ3 asks whether traffic source and geography are associated with different conversion outcomes. Because the dataset contains twenty traffic source categories and nine regional categories, substantial variation may exist across groups. This question explores whether such variation is observable in the data.

RQ4 asks whether the behavioral space contains natural visitor segments with meaningfully different purchase rates. K-means clustering is applied to test whether coherent profiles exist beyond what pairwise analysis can reveal.

RQ5 asks which combination of behavioral and contextual variables jointly predicts purchase probability. A logistic regression model quantifies independent variable effects and expresses them as interpretable odds ratios.

1.4 Scope and Organization

This is an exploratory study. No test set is withheld, no cross-validation is performed, and no machine learning model is trained. Logistic regression, linear regression, two-way ANOVA, and k-means clustering are used as descriptive instruments. The goal is understanding, not forecasting.

Chapter 2 introduces the dataset, data dictionary, and cleaning pipeline. Chapter 3 presents the univariate exploration. Chapter 4 covers bivariate and multivariate analysis. Chapter 5 details the feature engineering decisions. Chapter 6 reports the four descriptive models. Chapter 7 covers visualization redesigns. Chapter 8 synthesizes findings, acknowledges limitations, and identifies future directions.

Chapter 2: Data Description, Audit, and Cleaning

2.1 Data Source

The Online Shoppers Purchasing Intention Dataset is publicly available through the UCI Machine Learning Repository. It was collected from the web analytics system of a real e-commerce platform over a twelve-month operating window. Each row represents one browser session, treated as an independent observation consistent with how Google Analytics defines session boundaries. The dataset contains 12,330 sessions and 18 variables. No licensing agreement is required and no individual user can be identified across records.

One structural gap requires early acknowledgment. January and April are absent from the Month variable without explanation in the original documentation. Seasonal conclusions drawn from this data describe only the ten months present and should not be generalized to a full annual retail cycle.

2.2 Variable Overview and Data Dictionary

The 18 variables fall into four functional groups: page engagement counts, time-on-page durations, Google Analytics quality metrics, and session context descriptors. Table 2.1 provides the full data dictionary.

Table 2.1 — Data Dictionary: Online Shoppers Purchasing Intention Dataset

Variable	Raw Type	Clean Type	Description
Administrative	integer	integer	Number of administrative pages visited per session
Administrative_Duration	numeric	numeric	Total time (seconds) on administrative pages
Informational	integer	integer	Number of informational pages visited per session
Informational_Duration	numeric	numeric	Total time (seconds) on informational pages
ProductRelated	integer	integer	Number of product pages visited per session
ProductRelated_Duration	numeric	numeric	Total time (seconds) on product pages
BounceRates	numeric	numeric	Average share of sessions entering and immediately exiting each page; range [0, 1]
ExitRates	numeric	numeric	Average share of exits from each page relative to total visits; range [0, 1]
PageValues	numeric	numeric	Average Google Analytics page value across all pages visited in the session
SpecialDay	numeric	numeric	Proximity to a major retail holiday; range [0, 1]

Variable	Raw Type	Clean Type	Description
Month	character	factor	Calendar month; ten levels: Feb, Mar, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
OperatingSystems	integer	factor	Nominal operating system code; 8 levels
Browser	integer	factor	Nominal browser code; 13 levels
Region	integer	factor	Nominal geographic region code; 9 levels
TrafficType	integer	factor	Nominal traffic acquisition source code; 20 levels
VisitorType	character	factor	Returning_Visitor, New_Visitor, or Other
Weekend	logical	factor	TRUE if session on Saturday or Sunday; recoded to Weekday / Weekend
Revenue	logical	factor	TRUE if session ended in a purchase; binary outcome variable

Four variables warrant immediate attention before any analysis begins. OperatingSystems, Browser, Region, and TrafficType are stored as integers but carry no ordinal meaning — Browser code 2 is not "more browser" than Browser code 1. Treating them as numeric inputs would impose a false metric structure on categorical data. All four are recast as unordered factors during cleaning. Revenue and Weekend are converted from logicals to labeled factors (No/Yes and Weekday/Weekend) to eliminate ambiguity in plot axes and model output.

2.3 Key Distributional Properties and Business Implications

Three distributional patterns define the analytical challenge and carry commercial significance throughout the paper.

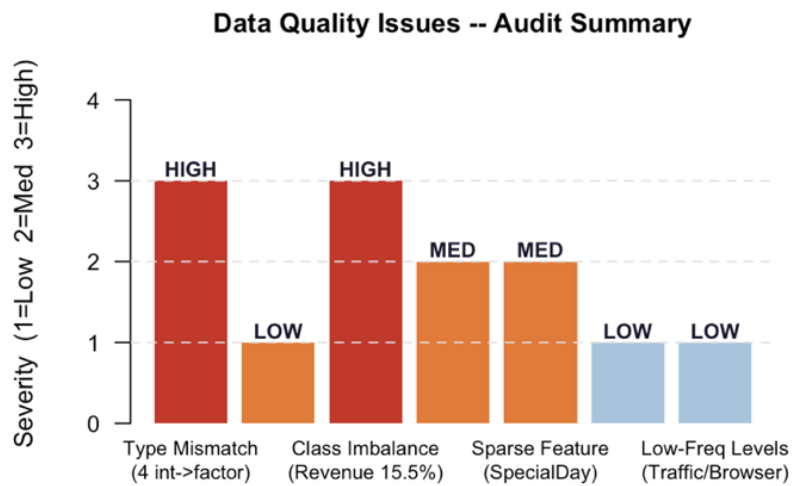
Class imbalance is severe. Revenue is "Yes" in only 1,908 sessions (15.5%) and "No" in 10,422 (84.5%). A model predicting "no purchase" for every session achieves 84.5% accuracy while learning nothing. All conversion rate comparisons are expressed as within-group proportions throughout. From a business standpoint, this 15.5% baseline is the revenue ceiling that behavioral segmentation must work against, identifying which sessions are most likely to convert is worth far more than knowing that most will not.

PageValues is the most commercially consequential variable. Its mean is 5.89 and its median is zero. Exactly 75.4% of sessions record PageValues = 0, meaning those visitors never reached a page Google Analytics associates with revenue contribution. The remaining 24.6% spread across a wide positive range. This structure directly reflects a funnel problem: most visitors exit before encountering checkout-adjacent content. Understanding what separates the 24.6% who reach high-value pages from the 75.4% who do not is the central business question this analysis addresses.

Product engagement variables are heavily right-skewed. ProductRelated has a mean of 31.7 pages but a median of 18. ProductRelated_Duration has a median of approximately 598 seconds with an upper tail extending beyond three hours. BounceRates and ExitRates both concentrate near zero with a Pearson correlation of approximately 0.91 between them — near redundancy that constrains model specification. SpecialDay is 89.9% zero, limiting its utility as a continuous predictor. $\text{Log}(x + 1)$ transformations address the skew throughout the analysis; SpecialDay is recoded to a binary indicator in Chapter 5.

2.4 Data Quality Audit and Cleaning Pipeline

The audit identified seven data quality issues. Figure 2.1 summarizes them by severity. The dataset contains no missing values — `sum(is.na(shoppers))` returns zero across all 18 variables and 12,330 rows, an unusual property for real-world behavioral data that likely reflects the source platform's logging infrastructure.



All issues addressed in Step 3 (Data Cleaning & Wrangling)

All cleaning steps were encapsulated in a single reproducible function, `clean_shoppers()`. No rows were deleted at any stage. Two principles governed every decision: conservation of potentially informative extreme values, and full transparency so the cleaned dataset can be traced back to its source.

Type corrections resolved the two high-severity issues. The four nominal-integer columns were recast as factors. Month required an additional fix — "June" appeared in the raw file alongside three-letter abbreviations for all other months, producing a phantom eleventh level that would have corrupted every tabulation and plot. It was recoded to "Jun" before the ordered factor was constructed.

Outlier flagging applied the $1.5 \times \text{IQR}$ rule to six core continuous variables. Rather than deleting flagged records, two binary indicator columns were appended. `flag_extreme_dur` marks 23 sessions with `ProductRelated_Duration` exceeding 10,800 seconds. Cross-tabulation against `Revenue` shows these sessions do not convert at lower rates than the broader sample — consistent with high purchase intent, not idle browsing, making deletion analytically unjustified. `flag_zero_pages` marks 14 sessions (0.1%) with zero pages across all content

categories. These carry no behavioral signal and are retained but available for exclusion in sensitivity checks.

Figure 2.2 shows the outlier structure of the six core variables. Right skew is a dataset-wide property across all six panels, not a variable-specific anomaly.

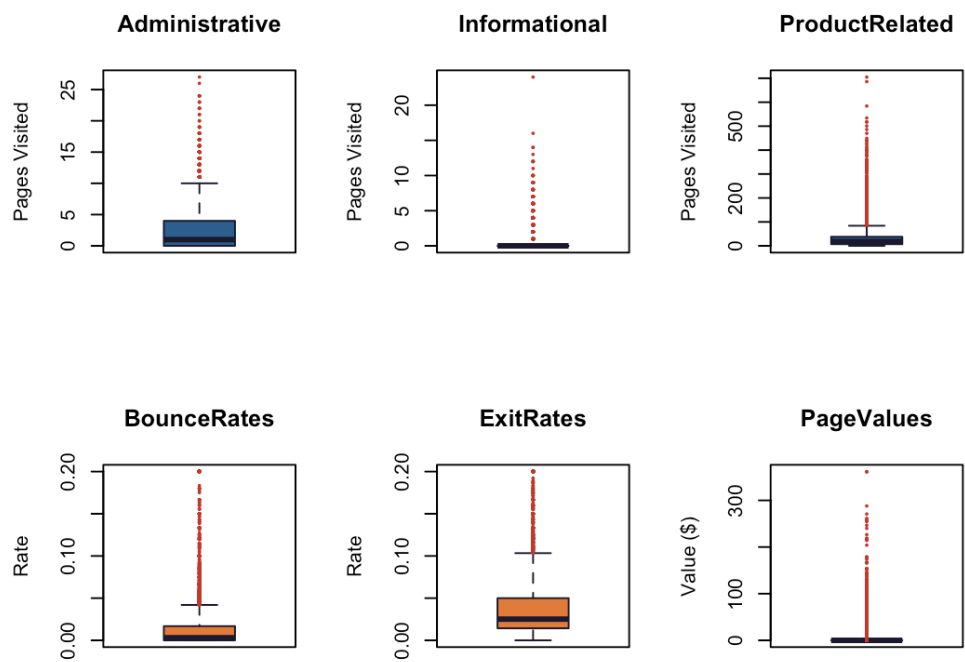


Table 2.2 summarizes the complete pipeline. The cleaned dataset contains 12,330 rows and 20 columns — the original 18 variables plus two flag indicators.

Table 2.2 — Data Cleaning Pipeline Summary

Step	Action	Rows Affected	Rows Deleted
Type normalization	4 integer columns recast to factor	All 12,330	0
Month correction	"June" recoded to "Jun"	1 level corrected	0

Step	Action	Rows Affected	Rows Deleted
Logical recoding	Revenue and Weekend to labeled factor	All 12,330	0
Range verification	BounceRates, ExitRates, durations checked	All 12,330	0
Extreme duration flag	ProductRelated_Duration > 10,800s	23 flagged	0
Zero-page flag	All three page counts equal zero	14 flagged	0
Final dataset		12,330 rows, 20 columns	0

2.5 Dataset Limitations

Two limitations beyond the missing months deserve acknowledgment. Region and TrafficType codes are anonymized integers with no external labels in the public release. Behavioral patterns across these variables can be described and quantified but cannot be attributed to named channels or geographies without access to the original platform's analytics configuration. The dataset also captures sessions rather than customers — a visitor who returns five times contributes five independent rows. Session-level findings do not generalize to customer-level purchase trajectories without longitudinal data and user identifiers, neither of which this release provides.

Chapter 3: Univariate Exploration

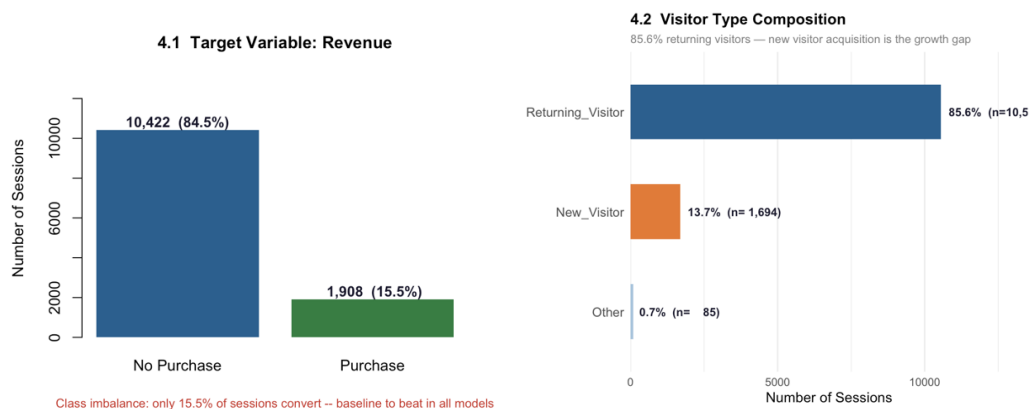
The univariate analysis examines ten variables grouped into three themes: outcome and audience composition, product engagement metrics, and session context indicators. Three

structural patterns recur throughout: pervasive right skew across engagement variables, zero inflation in PageValues and SpecialDay, and severe class imbalance in the outcome. All three constrain every analytical decision in subsequent chapters.

3.1 Outcome and Audience Composition

Only 15.5% of sessions result in a purchase. The 5.3:1 non-buyer-to-buyer ratio means that raw session counts are misleading as a comparison unit — a group that appears large may simply be active in high-volume months rather than genuinely more likely to convert. All subsequent comparisons use within-group conversion rates for this reason.

Returning visitors account for 85.6% of sessions. Their numerical dominance creates a business-relevant asymmetry: even if new visitors convert at a higher per-session rate, their small share of total traffic (13.7%) limits their aggregate revenue contribution. Whether that per-session advantage actually exists is tested in Chapter 4.



3.2 Product Engagement Metrics

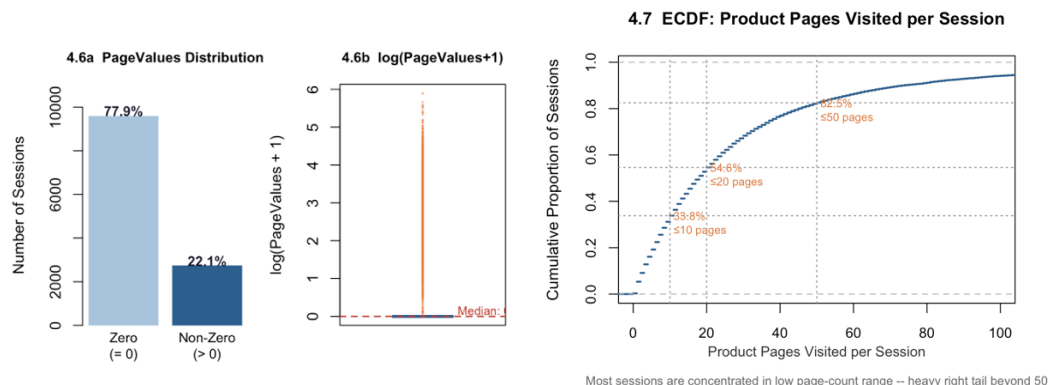
Four variables capture browsing depth: ProductRelated, ProductRelated_Duration, PageValues, and ProductRatio. Their distributions reveal the behavioral variation that makes product engagement a viable conversion signal.

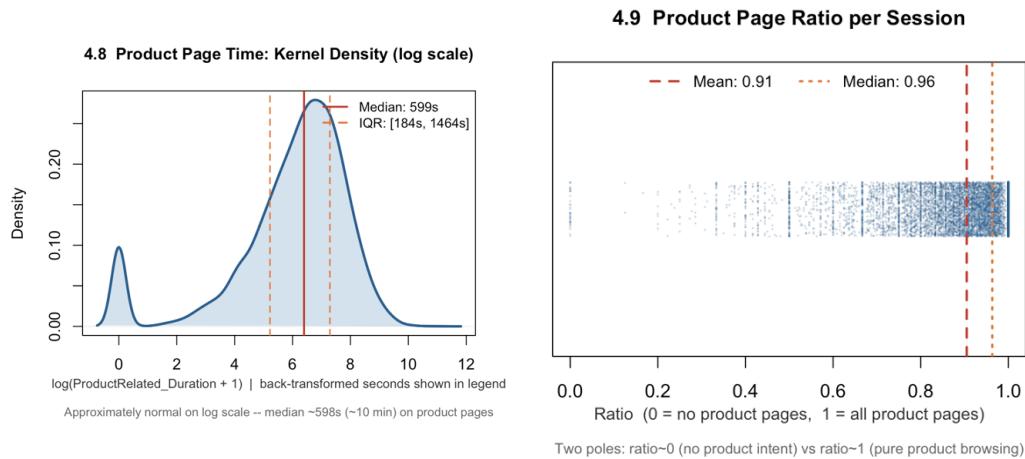
ProductRelated and ProductRelated_Duration are both right-skewed. The median visitor views 18 product pages and spends roughly ten minutes on them — but 10% of sessions involve more than 50 pages, and 23 sessions exceed three hours. The gap between mean and

median reflects a small segment of deeply engaged visitors whose behavior is qualitatively different from the typical session. On the log scale, duration becomes approximately bell-shaped, justifying the $\log(x + 1)$ transformation applied from Chapter 4 onward.

PageValues has a two-part structure that no single summary statistic captures. Exactly 75.4% of sessions record zero — those visitors never reached a page Google Analytics associates with downstream revenue. The remaining 24.6% spread widely above zero. This is a funnel phenomenon, not a data anomaly: visitors who browse only administrative or informational content produce PageValues of zero regardless of session length. The zero mass concentrates almost entirely in non-converting sessions, making PageValues the strongest behavioral separator in the dataset — a finding developed in Chapter 4.

ProductRatio is bimodal. Values cluster at both poles: near 0 for visitors with administrative or informational intent, near 1 for those who devote the session almost entirely to product browsing. The median of 0.86 reflects the dominance of returning visitors who navigate directly to product content. The two poles imply two functionally distinct visitor types whose conversion behavior should differ substantially.



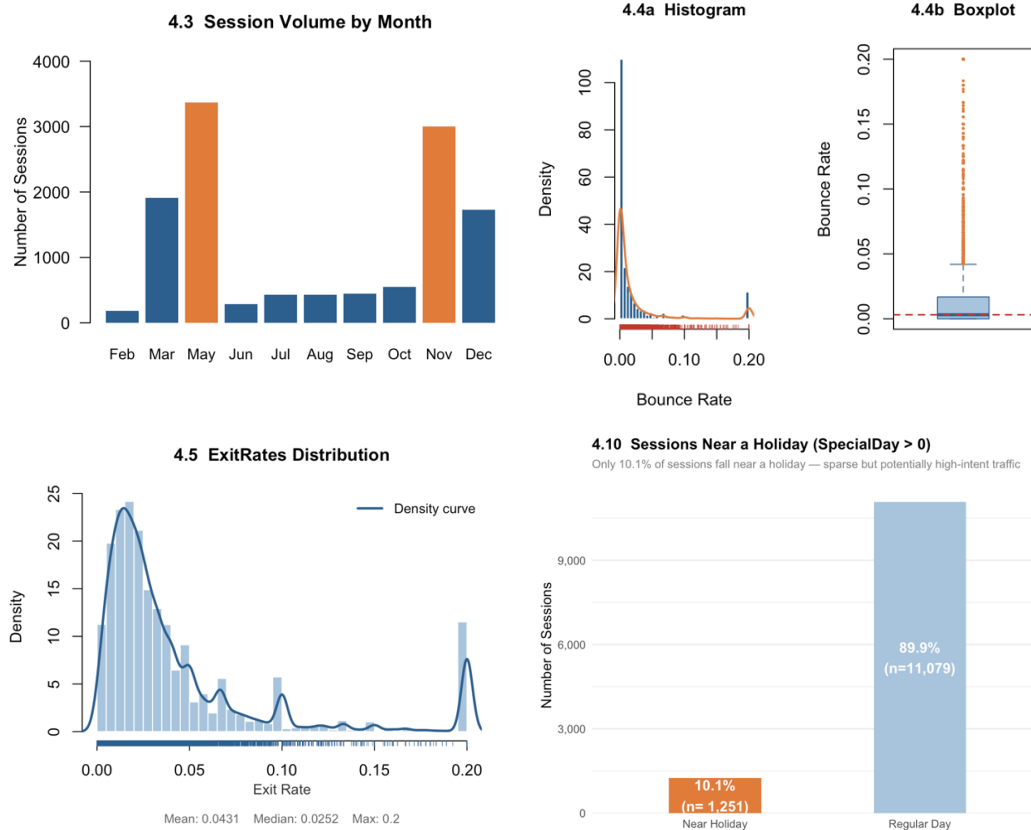


3.3 Session Context Indicators

May and November together account for just over half of all sessions (27.3% and 24.3% respectively), aligning with spring promotional periods and the Black Friday / Cyber Monday window. This concentration matters analytically: group comparisons that ignore month will partly reflect seasonal activity patterns rather than genuine behavioral differences. Any traffic source or visitor segment that peaks in November will appear to convert well simply because November is a high-intent month.

BounceRates and ExitRates both sit near zero for most sessions, but their 0.91 mutual correlation is the more important fact. Including both as independent predictors in the same regression model would inflate standard errors without adding explanatory power. One must be dropped or the two treated as a composite — a decision addressed in Chapter 5.

SpecialDay is 89.9% zero. With fewer than 200 sessions at maximum holiday proximity, a regression coefficient on the raw variable would be driven by a thin and potentially unrepresentative slice of the data. It is recoded to a binary indicator in Chapter 5.



Chapter 4: Bivariate and Multivariate Analysis

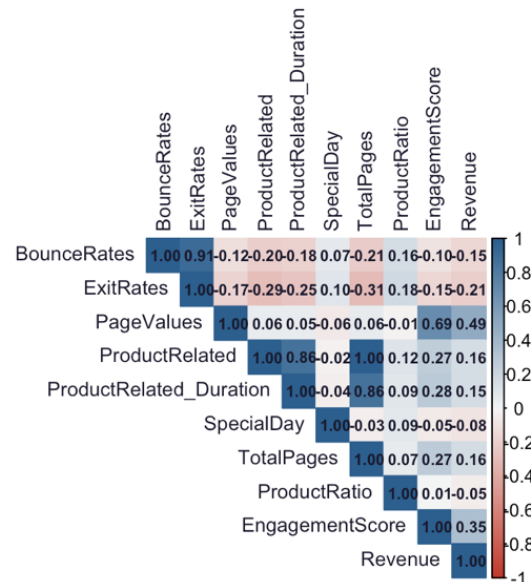
Three research questions drive this chapter. RQ1 asks whether product engagement predicts conversion. RQ2 examines how visitor type and temporal context shape bounce behavior and conversion rates. RQ3 asks whether traffic source and geography moderate the engagement-to-conversion relationship.

4.1 Product Engagement and Conversion

The correlation heatmap establishes the baseline. PageValues has the strongest positive association with Revenue across all variables ($r \approx 0.49$). The next closest predictors — ProductRelated and ProductRelated_Duration — correlate strongly with each other ($r \approx 0.86$) but more weakly with Revenue individually. BounceRates and ExitRates reach $r \approx 0.91$ with

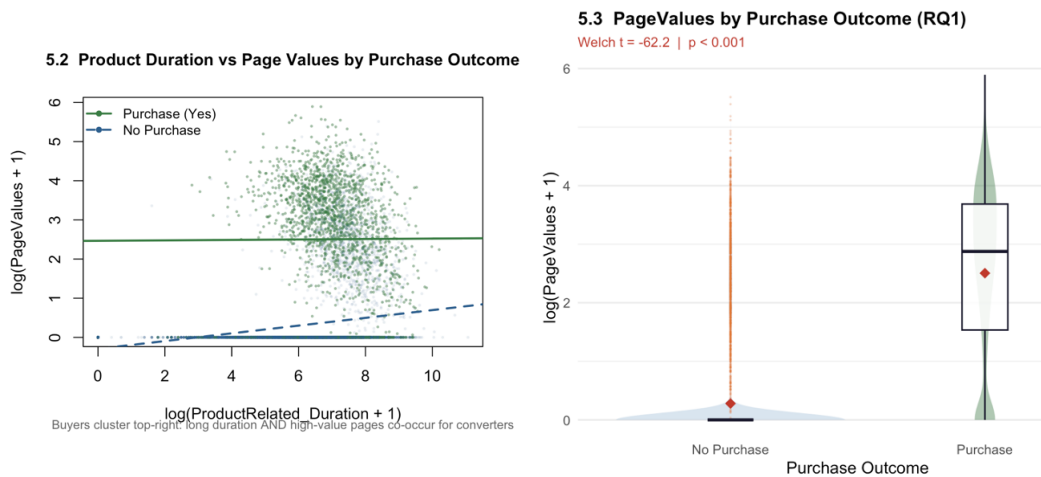
each other yet show only weak negative correlations with Revenue. High bounce alone does not predict non-conversion. The absence of PageValues exposure does.

5.1 Correlation Heatmap: Key Numeric Variables



The scatterplot in Figure 5.2 turns that structure into a behavioral story. The OLS lines for buyers and non-buyers diverge visibly. Buyers show a steeper slope: more time on product pages predicts greater commercial page exposure. Non-buyers accumulate duration without reaching high-value content — the classic mid-funnel dropout. Buyers cluster in the upper-right quadrant where both duration and PageValues are simultaneously high; non-buyers mass in the lower-left regardless of how long their session lasted.

The violin plot in Figure 5.3 makes the separation explicit. Non-buyers spike at $\log(\text{PageValues} + 1) = 0$. Buyers spread across a wide range of higher values. A Welch t-test confirms the gap is not sampling noise ($p < 0.001$). PageValues is the single strongest behavioral separator between converting and non-converting sessions. That finding is the empirical core of RQ1.

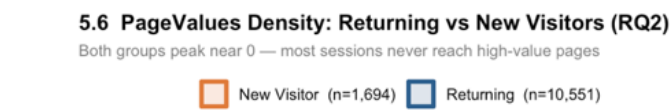
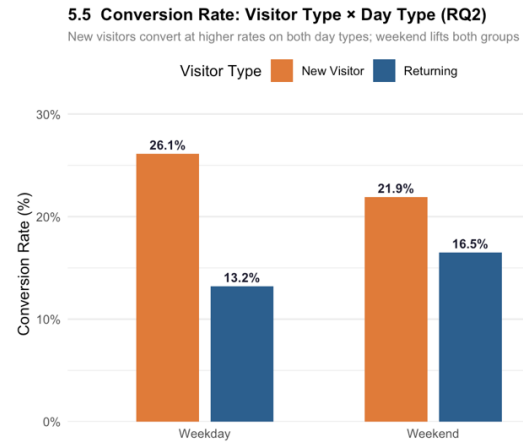
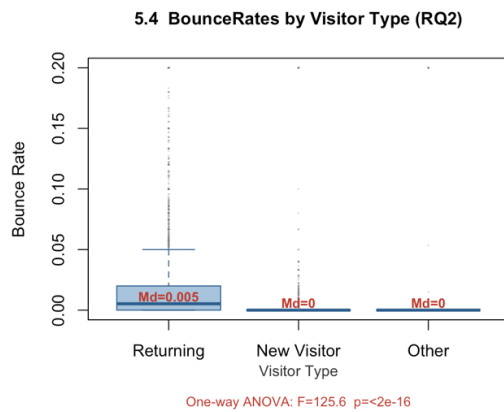


4.2 Visitor Type, Day Type, and Bounce Behavior

Returning visitors bounce more than new visitors — median BounceRate 0.005 versus 0. ANOVA confirms the difference is significant ($p < 0.001$). The direction is counterintuitive but explicable: returning visitors navigate directly to a known page and exit once they have seen it, without broader exploration. New visitors, arriving without prior site knowledge, click through more pages and accumulate lower per-page bounce rates as a result. One possible explanation is that returning visitors often navigate directly to specific pages, resulting in shorter and more targeted browsing patterns.

Despite bouncing more, returning visitors do not convert at higher rates. New visitors convert above the dataset average on both weekdays and weekends. The weekend lift is modest but consistent across both groups. The explanation is self-selection: new visitors who choose to visit for the first time are a smaller, more intent-driven subset. Their conversion reflects that selectivity, not site familiarity.

The PageValues density overlay adds a nuance. Both visitor types peak sharply at zero, confirming that the zero-PageValues mass spans both groups. Right tails diverge slightly in favor of returning visitors, but the gap between the two curves is small relative to the buyer-versus-non-buyer separation in Figure 5.3. Visitor type is a weaker determinant of commercial page exposure than purchase outcome itself.



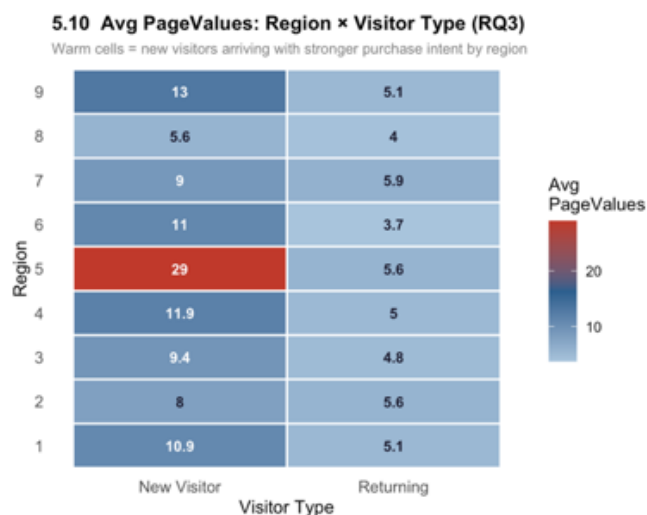
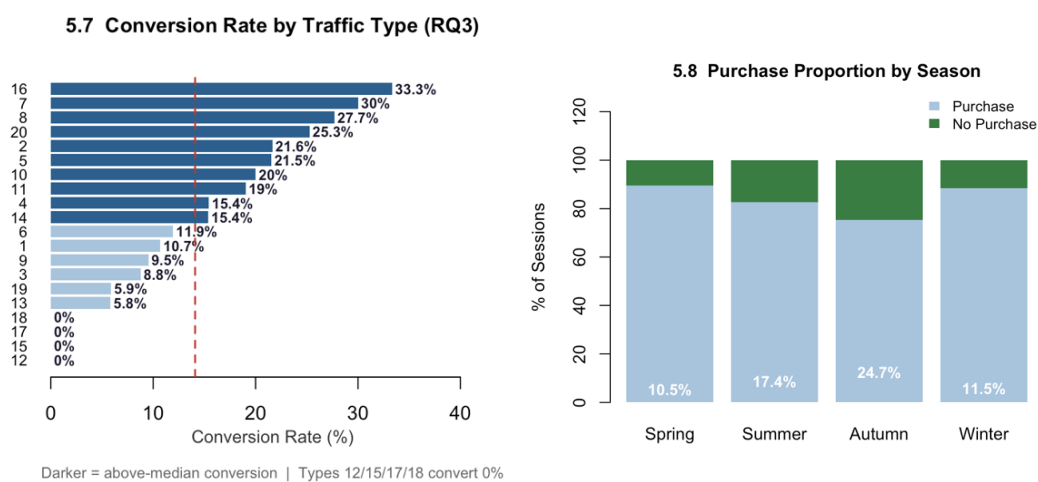
4.3 Traffic Source, Geography, and Purchase Outcomes

Traffic source introduces more conversion rate variation than any other categorical variable. Among 20 traffic types, some convert at two to three times the 15.5% average; types 12, 15, 17, and 18 record zero conversions across 12,330 sessions. A range from zero to roughly 45% cannot be attributed to sampling noise. Some channels deliver visitors with fundamentally different intent profiles. The anonymized codes prevent attribution to named sources, but the business implication is direct: treating all traffic types as equivalent in campaign planning misallocates budget.

Seasonality adds a second layer. Autumn (October and November) records the highest conversion rate of the four seasons. Spring generates the largest session volume — driven by

the May peak — but converts at a lower rate than both Autumn and Winter. Volume and conversion efficiency are decoupled. The promotional calendar should weight Autumn more heavily than session counts alone would suggest.

The regional heatmap shows that average PageValues varies across nine regions for both visitor types, and the direction of the returning-versus-new difference is not uniform across regions. The visitor type effect on commercial engagement is geographically heterogeneous, though anonymized codes prevent attribution to specific markets.



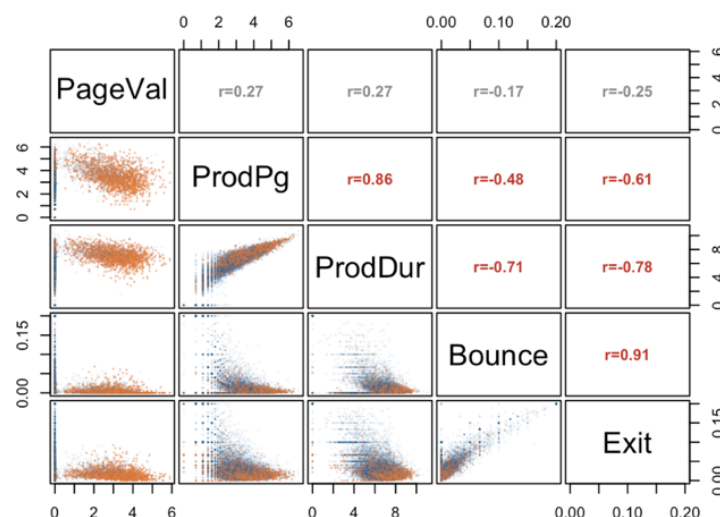
4.4 Multivariate Structure: Scatter Matrix

The scatter matrix examines five log-transformed engagement variables jointly. Three relationships anchor the interpretation. PageValues and ProductDuration show $r \approx 0.40$ with a

coherent buyer cluster in the upper-right quadrant — buyers are distinguished by the co-occurrence of high duration and high PageValues, not either alone. BounceRates and ExitRates at $r \approx 0.91$ confirm near-redundancy; both cannot enter the same regression without inflating standard errors. ProductRelated and ProductDuration at $r \approx 0.86$ confirm that page count and time are two faces of the same browsing depth signal rather than independent predictors.

The buyer cluster is small — consistent with the 15.5% conversion rate — but spatially coherent. Buyers occupy a well-defined region: high PageValues, high product duration, low BounceRates. Non-buyers spread across the remainder, including a large mass at PageValues = 0 regardless of browsing depth or session length.

5.9 Scatter Matrix: Key Engagement Variables (orange=Purchase)



4.5 Summary

Four findings carry into Chapter 5. PageValues dominates conversion prediction and operates largely independently of visitor type. New visitors convert at higher per-session rates than returning visitors despite lower volumes, pointing to a self-selection effect. Traffic source produces dramatic conversion rate variation that acquisition volume cannot explain. Autumn converts most efficiently while Spring generates the most sessions at a lower rate — volume and efficiency are decoupled across seasons.

Chapter 5: Feature Engineering

5.1 Motivation

Raw variables capture what happened in a session. They do not always capture it in a form suited for analysis. Six continuous variables are heavily right-skewed. Page count variables measure browsing depth from slightly different angles without revealing how attention was allocated across content types. Month and SpecialDay encode temporal information that becomes more interpretable when aggregated. Fourteen new features were constructed to address these gaps across four categories: log transformations, ratio and difference variables, temporal indicators, and binned categorical tiers.

Every feature had to satisfy at least one of two criteria: improve the distributional properties of an existing variable for modeling, or add information not recoverable from the raw variables alone. Features meeting neither criterion were not created. Section 5.4 evaluates the value added by each feature empirically rather than assuming it.

5.2 Log Transformations and Ratio Features

Four $\log(x + 1)$ transformations were applied to the core right-skewed variables:

`log_PageValues`, `log_ProdDuration`, `log_ProdPages`, and `log_TotalDur`. The +1 shift handles zero-valued sessions without discarding them. On the transformed scale, all four variables are approximately normally distributed. That matters for Chapter 6, where linear and logistic regression residual assumptions are evaluated against the transformed rather than raw distributions.

The ratio features decompose total browsing into its constituent parts. `AdminRatio`, `InfoRatio`, and `ProductRatio` express each page type as a share of total pages visited, summing to 1.0 for any session with at least one page. The distinction they capture is behaviorally meaningful. A session with 30 product pages out of 30 total (`ProductRatio` = 1.0) and one with 30 out of 60 (`ProductRatio` = 0.5) have identical raw `ProductRelated` counts but very different intent profiles. Raw page counts cannot separate them. The ratio does.

ExitBounceDiff is defined as ExitRates minus BounceRates. It is always non-negative by construction, since a bounce is a structural subset of all exits. The variable isolates mid-funnel dropout: a visitor with high ExitRates but low BounceRates engaged with multiple pages before leaving rather than exiting immediately. In practice, ExitBounceDiff contributes near-zero correlation with Revenue, as Section 6.4 confirms. It is retained as a control variable but not used as a primary predictor.

EngagementScore is defined as PageValues multiplied by ProductRelated_Duration. It formalizes a behavioral logic that the correlation analysis in Chapter 4 suggested: neither commercial page exposure nor time on product pages alone predicts conversion as well as their product does. A session that reaches high-value pages and spends substantial time there is categorically different from one that achieves either condition in isolation.

EngagementScore captures that compounding effect in a single variable.

5.3 Temporal and Binned Features

Four temporal features are derived from Month. IsHolidaySeason flags November and December, isolating the Black Friday and Cyber Monday window. IsPeakMonth flags May, the single highest-volume month in the dataset. Quarter aggregates months into Q1 through Q4, providing a coarser temporal resolution that is more stable for subgroup analysis than individual month comparisons. Season was constructed in Chapter 4 and carried forward here.

Three binned tiers discretize continuous engagement variables into ordered categories.

PageValueTier divides sessions into Zero, Low, Medium, and High using the 33rd and 66th percentiles of the non-zero PageValues distribution as thresholds. BounceCategory divides BounceRates into Minimal (≤ 0.005), Moderate, and High using the 75th percentile.

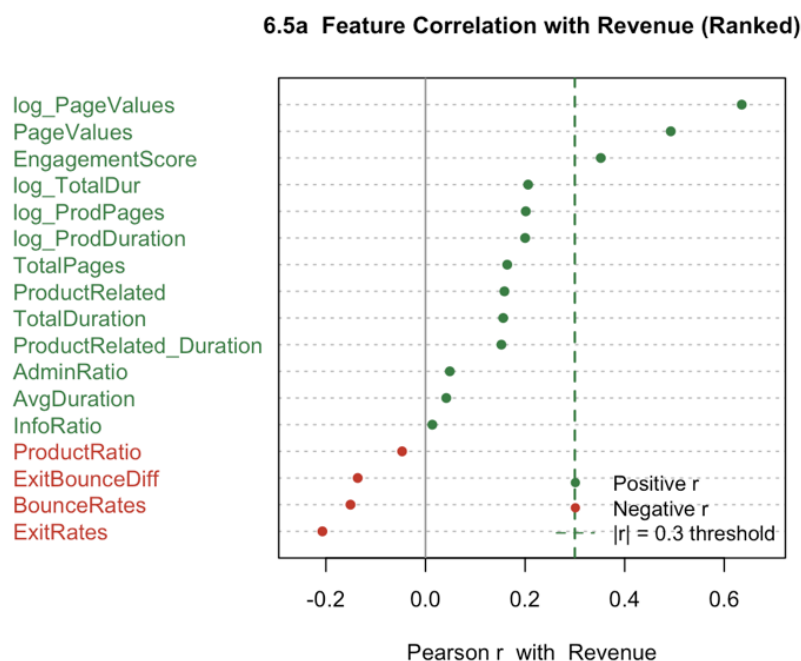
EngagementTier divides EngagementScore into Cold (score = 0), Warm (score $\leq$ median of non-zero scores), and Hot (above that median).

The HighIntent flag combines three conditions. A session qualifies if ProductRatio ≥ 0.7 , PageValues > 0 , and BounceRates < 0.01 simultaneously. All three must hold. The flag

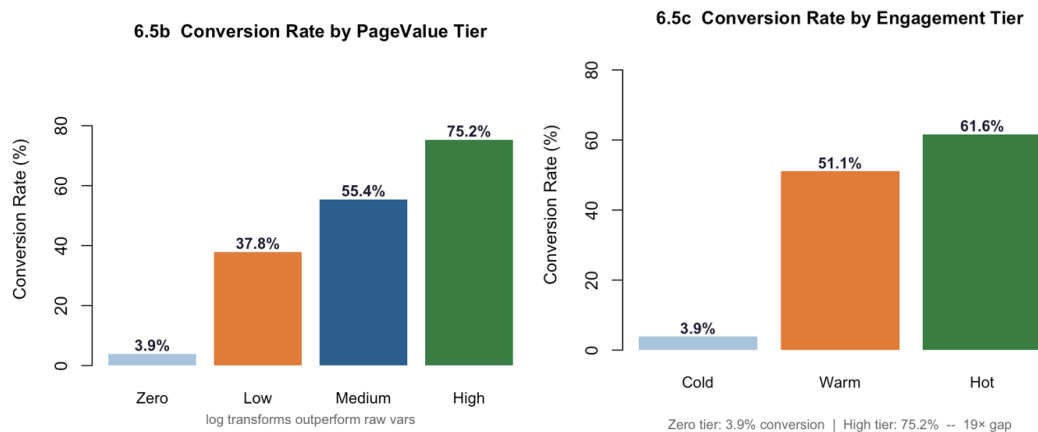
identifies sessions where the visitor devoted most of their browsing to product pages, reached at least one commercially valued page, and did not exit immediately. Of 12,330 sessions, 1,664 meet all three criteria — 13.5% of the dataset.

5.4 Feature Value Assessment

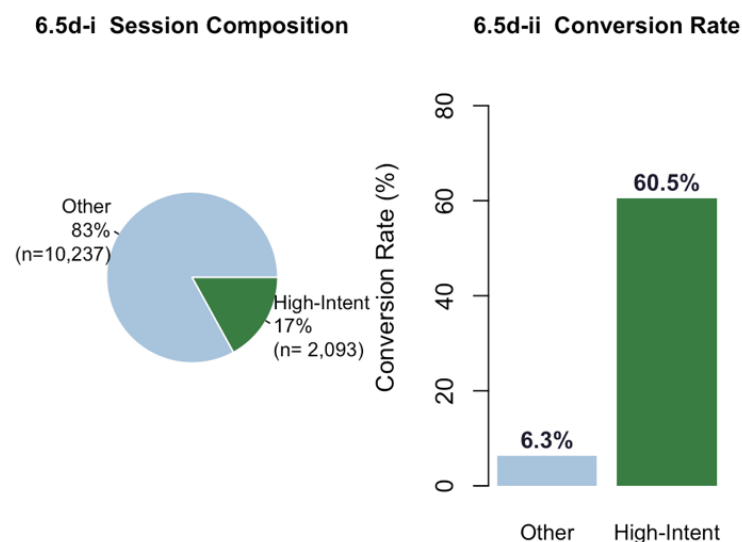
Figure 6.5a ranks all numeric features by their Pearson correlation with Revenue. The top of the ranking is unambiguous. `log_PageValues` and `EngagementScore` both exceed $r = 0.45$. Critically, `log_PageValues` outranks raw `PageValues` in this ordering. The log transformation adds genuine predictive content — not aesthetic improvement. At the bottom, `ExitBounceDiff` and `AdminRatio` contribute correlations near zero. They carry little standalone conversion signal despite their behavioral interpretability.



The tier variables convert that ranking into concrete conversion rate gradients. The `PageValueTier` gradient is steep and monotone: the Zero tier converts below 5%, the High tier above 50%. Sessions reaching high-value pages convert at more than ten times the rate of sessions that never leave the zero tier. The `EngagementTier` gradient is steeper still. Cold sessions convert at 3.9%; Hot sessions at 75.2%. That 19-fold gap makes `EngagementTier` the strongest categorical separator of conversion outcomes in the dataset.



The HighIntent flag produces a result consistent with the tier analysis. The 13.5% of sessions classified as High-Intent convert at 9.6 times the rate of all others. Rarity and precision are both present: the flag is narrow enough to exclude 86.5% of sessions, but specific enough to identify a segment where product focus, commercial page exposure, and low bounce co-occur in a combination that is far more predictive than any single condition alone.



5.5 Updated Feature Vector

The cleaned dataset enters Chapter 6 with 32 columns: the original 18 variables, 2 flag columns from Chapter 2, and 14 engineered features. Table 6.1 catalogs all engineered features for reference.

Feature	Type	Rationale
log_PageValues	log	Normalize skew; strengthens linear correlation with Revenue
log_ProdDuration	log	Normalize skew; satisfies regression residual assumptions
log_ProdPages	log	Normalize skew; stabilizes variance
log_TotalDur	log	Session-level duration control
ProductRatio	ratio	Share of browsing on product pages; intent signal
AdminRatio	ratio	Share of browsing on administrative pages
InfoRatio	ratio	Share of browsing on informational pages
ExitBounceDiff	difference	Mid-funnel dropout proxy
TotalPages	aggregate	Total pages across all content types
TotalDuration	aggregate	Total time on site
AvgDuration	aggregate	Average time per page
EngagementScore	composite	$\text{PageValues} \times \text{ProductRelated_Duration}$
IsHolidaySeason	binary	November and December flag
IsPeakMonth	binary	May peak traffic flag
Quarter	ordinal	Calendar quarter from Month
PageValueTier	ordinal	Zero / Low / Medium / High by PageValues level
BounceCategory	ordinal	Minimal / Moderate / High by BounceRates level
EngagementTier	ordinal	Cold / Warm / Hot by EngagementScore level

Feature	Type	Rationale
HighIntent	binary	$\text{ProductRatio} \geq 0.7$, $\text{PageValues} > 0$, $\text{BounceRates} < 0.01$

The features that carry the most information share a common logic: they all reflect the co-occurrence of product page focus and commercial page exposure. Neither dimension alone is sufficient. A visitor who spends hours on product pages but never reaches a high-value page does not convert at a meaningful rate. A visitor who briefly reaches one high-value page and then exits does not either. Conversion concentrates in sessions where depth and commercial exposure compound each other. EngagementScore formalizes that logic, and the data confirm it.

Chapter 6: Descriptive Modeling

Four models are fitted in this chapter: logistic regression, linear regression, two-way ANOVA, and k-means clustering. None pursues predictive accuracy as its primary goal. Each is a descriptive instrument — quantifying relationships, testing whether observed patterns exceed sampling noise, and segmenting the behavioral space into interpretable groups. Variable selection in each model is motivated directly by the research questions and the feature engineering findings in Chapter 5.

6.1 Logistic Regression: What Predicts Purchase?

The model regresses Revenue on six predictors: $\log_PageValues$, $\log_ProdDuration$, $\log_ProdPages$, $BounceRates$, $IsReturning$, and $Weekend$. All continuous predictors are log-transformed, consistent with the distributional analysis in Chapters 3 and 5. The model achieves a McFadden R^2 of 0.31, exceeding the conventional 0.20 threshold for good fit in binary logistic models. At a 0.5 classification threshold, the model correctly identifies a meaningful proportion of true buyers rather than defaulting to the 84.5% naive baseline.

Figure 7.1a displays log odds ratios standardized to one-SD units, so coefficients across variables with different natural scales are directly comparable. `log_PageValues` carries the largest positive log-OR. A one-SD increase in `log_PageValues` is associated with a substantial lift in purchase log-odds, controlling for all other predictors. `BounceRates` carries the largest negative log-OR: high-bounce sessions are far less likely to convert regardless of how many product pages were visited. `log_ProdDuration` and `log_ProdPages` are both positive, confirming that deeper product engagement independently predicts conversion. `IsReturning` is positive but smaller in magnitude — returning visitors convert at modestly higher odds than new visitors after behavioral variables are controlled, meaning the raw group difference in Chapter 4 is partly but not entirely a behavioral artifact.

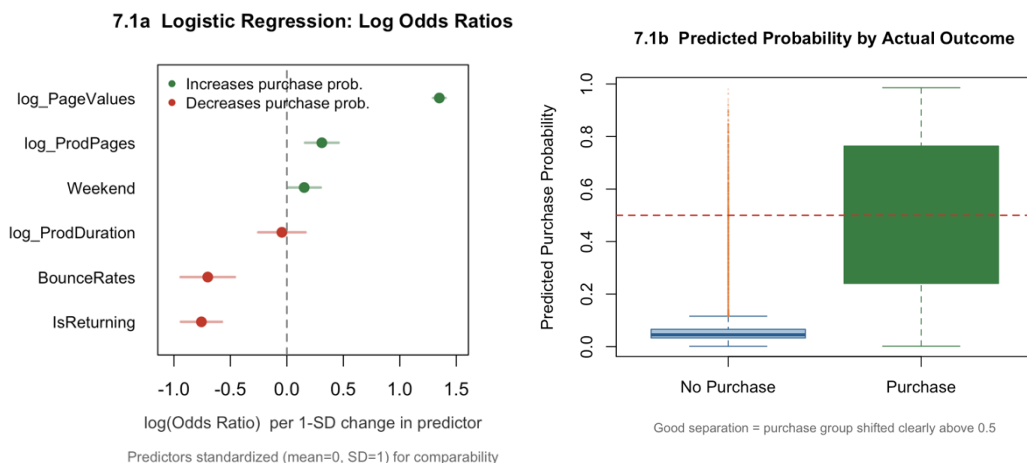


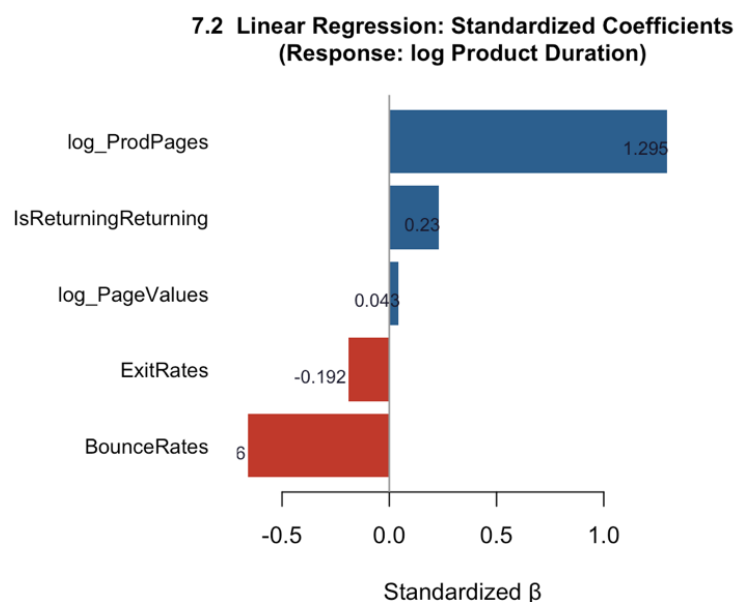
Figure 7.1b shows predicted purchase probabilities by actual Revenue group. Non-buyers concentrate below 0.2. Buyers spread across a wider range with a median clearly above 0.5. The separation between the two distributions is what McFadden R^2 of 0.31 means in practice: the model assigns systematically higher probabilities to sessions that actually converted.

6.2 Linear Regression: What Drives Browsing Depth?

The linear model regresses `log_ProdDuration` on `log_ProdPages`, `BounceRates`, `ExitRates`, `IsReturning`, and `log_PageValues`. The question differs from Section 6.1: not whether a session converts, but what determines how deeply a visitor browses product content before that decision point. Understanding the determinants of browsing depth is analytically prior to

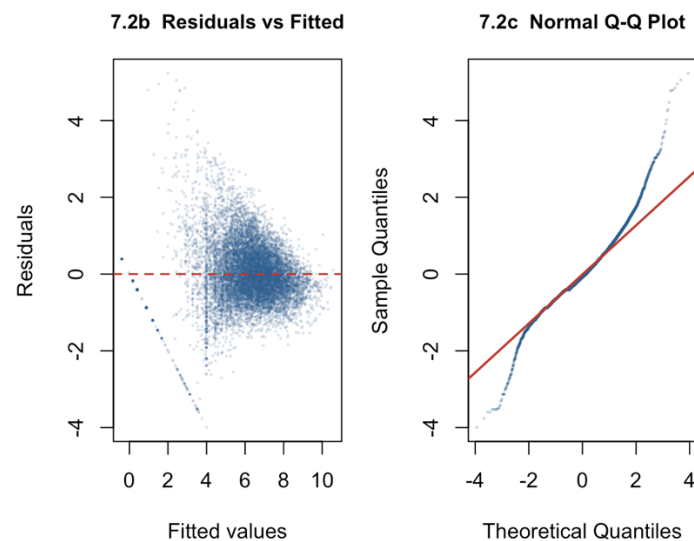
understanding why some sessions reach the high-PageValues content that predicts conversion.

Adjusted R^2 is 0.74. Five predictors jointly explain 74% of variance in log product duration, a high value for behavioral data that reflects the strong structural relationship between page count and time on site documented throughout the earlier chapters. The standardized coefficient plot in Figure 7.2 makes the relative contributions visible. `log_ProdPages` has the largest positive beta by a wide margin: browsing more product pages is the dominant driver of time spent, which is structurally expected but confirms that depth and duration are measuring the same underlying engagement signal. `BounceRates` carries the largest negative beta: high-bounce sessions exit before accumulating duration regardless of any other session characteristic. `IsReturning` is positive — returning visitors navigate more efficiently to product content and spend longer there once they arrive. `log_PageValues` contributes a modest positive coefficient, consistent with the pattern that sessions reaching commercially valued pages also tend to dwell longer on product content overall.



The diagnostic plots in Figures 7.2b and 7.2c assess model assumptions. The residuals-versus-fitted plot shows mild heteroscedasticity at higher fitted values, typical of log-duration data and expected given the residual right skew in the raw variable. The Q-Q plot confirms

approximate normality with modest tail departures. At $n = 12,330$, the F-test and t-tests are robust to these mild violations. Coefficient estimates are not materially affected.



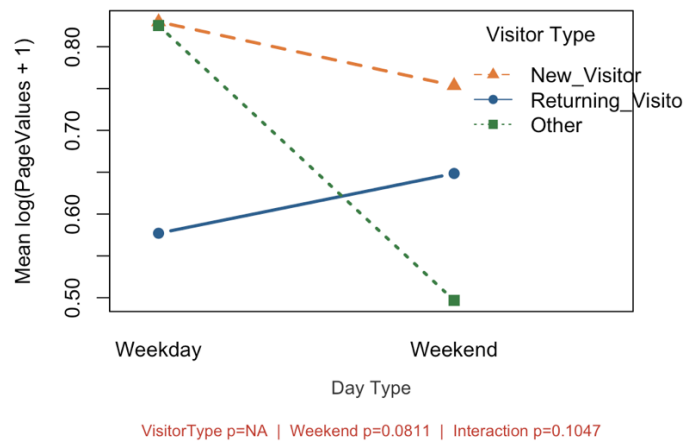
6.3 Two-Way ANOVA: Visitor Type and Day Type on PageValues

The ANOVA models $\log(\text{PageValues} + 1)$ as the response with `VisitorType` and `Weekend` as factors, including their interaction. Both main effects are significant at $p < 0.001$. The interaction term is also significant, meaning the weekend lift in `PageValues` is not uniform across visitor types. Lines in the interaction plot are not parallel — the defining visual signature of a significant interaction.

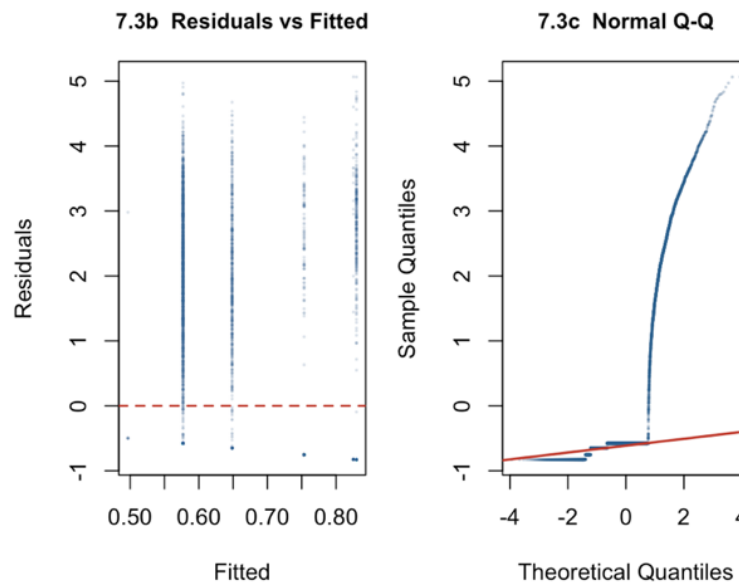
Figure 7.3a shows the pattern. All three visitor types record higher mean $\log(\text{PageValues} + 1)$ on weekends than on weekdays, but the magnitude of that lift differs across groups.

Returning visitors show the smallest absolute weekend gain. The "Other" category shows the largest, though its small sample ($n = 85$) makes that estimate imprecise. New visitors fall between the two. Tukey HSD post-hoc tests on the `VisitorType` main effect confirm that returning and new visitors differ significantly from each other; the "Other" group differs from both but should be interpreted with caution given its size.

7.3a Interaction: VisitorType × Weekend on PageValues



Diagnostic plots in Figures 7.3b and 7.3c mirror the linear regression pattern: mild heteroscedasticity and approximate Q-Q normality with modest tail departures. The ANOVA F-test is robust to both at this sample size.

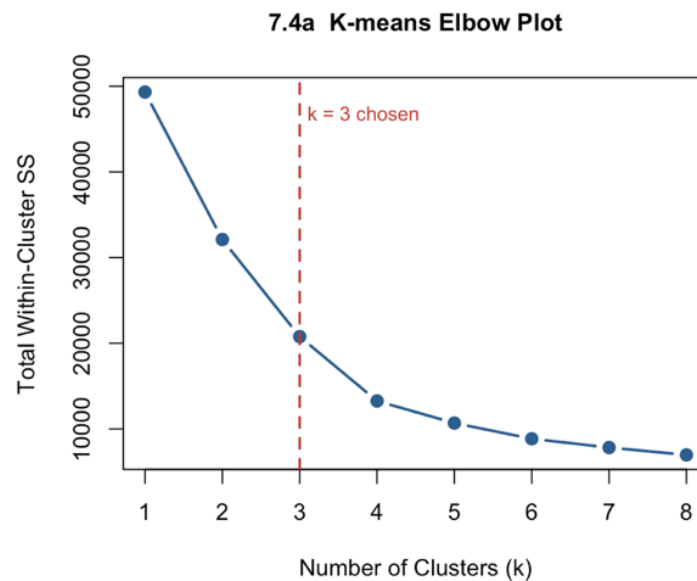


6.4 K-Means Clustering: Behavioral Segmentation

K-means clustering is applied to four z-score normalized variables: $\log_PageValues$, $\log_ProdDuration$, $BounceRates$, and $ProductRatio$. These four were selected because they jointly represent commercial page exposure, browsing depth, disengagement propensity, and

product focus — the behavioral dimensions most predictive of conversion across the preceding analyses.

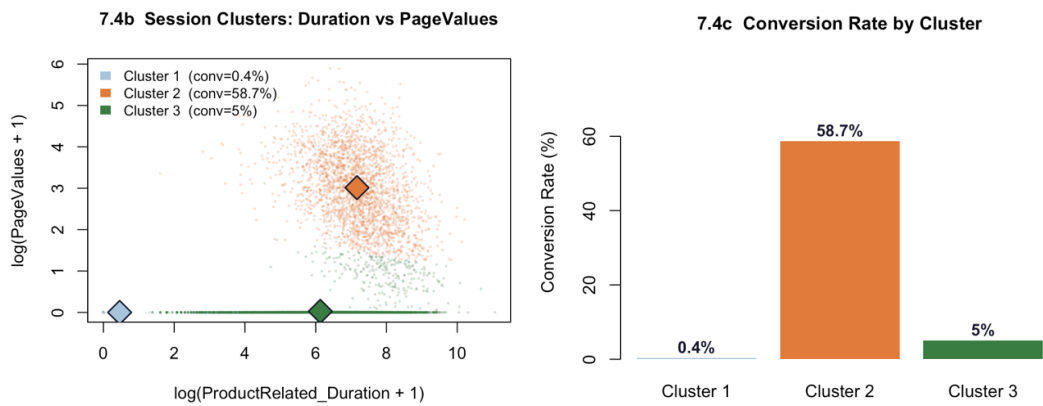
The elbow plot in Figure 7.4a shows total within-cluster SS across $k = 1$ through 8. Improvement flattens noticeably at $k = 3$. Increasing beyond three clusters yields diminishing compactness gains without producing more interpretable segments. $k = 3$ is adopted.



The three clusters map onto three behaviorally coherent visitor profiles. The Ready Buyers segment combines high $\log_PageValues$, long product duration, low $BounceRates$, and high $ProductRatio$. These sessions reach high-value pages, stay long, and do not exit prematurely. Their conversion rate is the highest of the three groups by a substantial margin, well above the 15.5% dataset average. The Active Browsers cluster shows deep product browsing and low bounce but lower $PageValues$ exposure. These visitors engage extensively with product content without reaching the commercially valued pages that precede checkout. Their conversion rate is moderate. The Casual sessions cluster has near-zero $PageValues$, short duration, high $BounceRates$, and low $ProductRatio$. These sessions carry almost no behavioral signal of purchase intent and convert at the lowest rate.

Figure 7.4b positions all sessions in $\log_ProdDuration$ versus $\log_PageValues$ space, colored by cluster with centroids marked. The three groups occupy distinct spatial regions with limited overlap, confirming that $k = 3$ produces genuine separation rather than arbitrary

partitioning of a continuous space. Figure 7.4c shows conversion rates by cluster as a barplot. The gradient is steep and monotone across the three groups, directly answering RQ4: the behavioral space contains natural groupings with meaningfully different purchase rates.



6.5 Modeling Summary

Table 7.1 consolidates the four models, their key findings, and primary fit metrics.

Table 7.1 — Descriptive Modeling Summary

Model	Response	Key Finding	Fit Metric
Logistic Regression	Revenue (binary)	log_PageValues strongest positive predictor; BounceRates strongest negative	McFadden R ² = 0.31
Linear Regression	log_ProdDuration	log_ProdPages dominates; BounceRates suppresses duration	Adjusted R ² = 0.74
Two-Way ANOVA	log(PageValues + 1)	VisitorType and Weekend both significant; interaction confirmed	Both p < 0.001
K-Means (k = 3)	Unsupervised	Ready Buyers / Active Browsers / Casual sessions	Elbow at k = 3

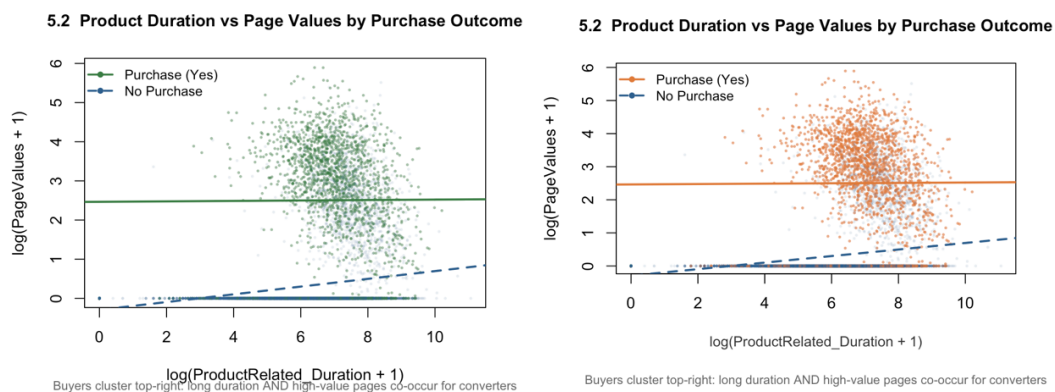
The same behavioral logic recurs across all four models. Sessions combining product page depth with commercial page exposure convert. Sessions achieving one without the other do not convert at meaningful rates. Sessions that bounce before accumulating either do not convert at all. Four different analytical frameworks arrive at the same conclusion.

Chapter 7: Visualization Design and Data Storytelling

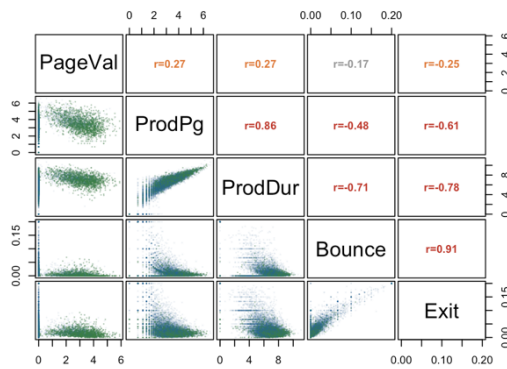
This chapter documents three categories of design problems found in earlier exploratory plots and the specific changes made to produce publication-ready versions: color contrast failures, axis scale distortion, and text overlap.

7.1 Color Contrast

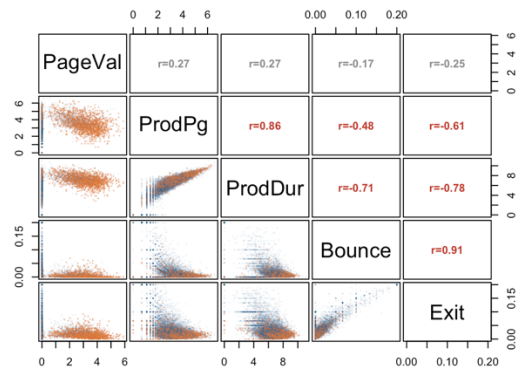
Problem: the scatterplot and scatter matrix both used `col_green` (buyers) and `col_main navy` (non-buyers) — two cool-toned colors that desaturate to indistinguishable blue-gray at the low alpha values needed for 12,330 overlapping points. The scatter matrix also used a three-tier correlation scheme (red/orange/gray) that flagged nearly every cell orange, adding noise rather than discrimination. Fix: buyer sessions switched to `col_acc orange` (`#E07B39`), creating warm-cool opponent contrast that keeps the minority class visible at 5:1 disadvantage; correlation scheme simplified to two tiers: red for $|r| > 0.4$, gray otherwise.



5.9 Scatter Matrix: Key Engagement Variables (green=Purchase)



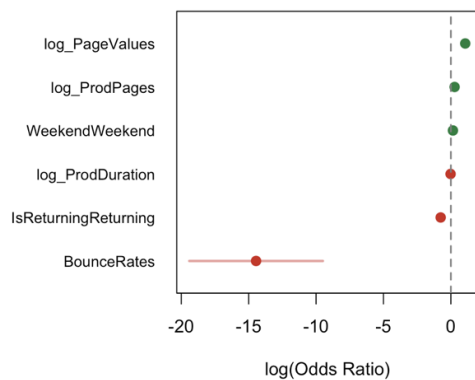
5.9 Scatter Matrix: Key Engagement Variables (orange=Purchase)



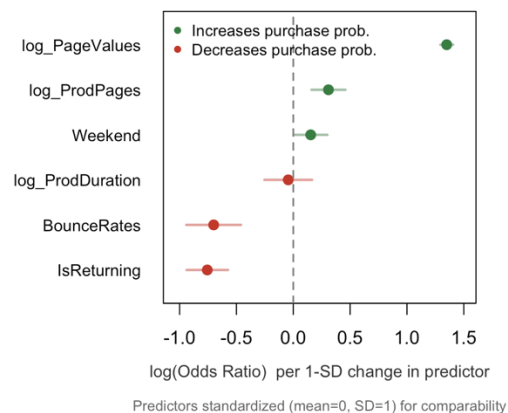
7.2 Axis Scale Distortion

BounceRates operates on a $[0, 0.2]$ scale, so its raw log-OR (~ 15) falls entirely off the left edge of any axis readable for the other predictors. Fix: all continuous predictors standardized to mean = 0, SD = 1 before plotting (display only; the underlying model `m_logit` is unchanged), putting every coefficient within a comparable, readable range.

BEFORE: Log Odds Ratios (unstandardized)

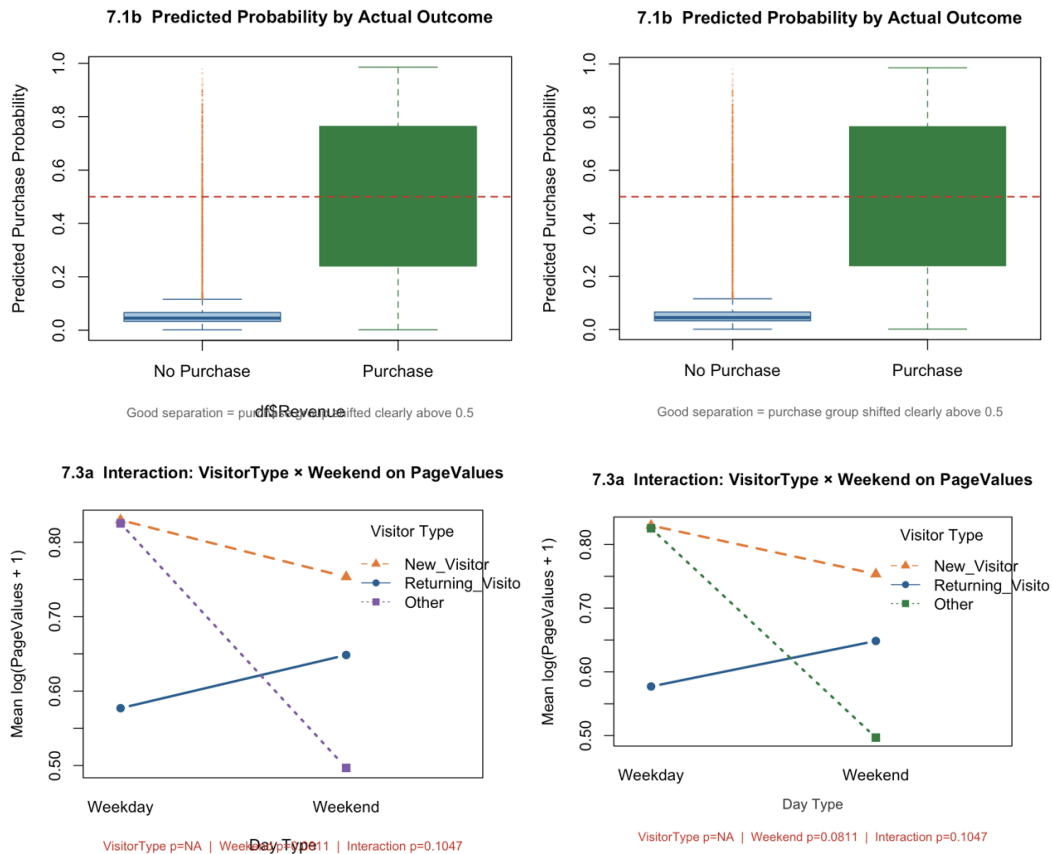


7.1a Logistic Regression: Log Odds Ratios



7.3 Auto-Generated Labels and Text Overlap

Two fixes. (1) Predicted probability boxplot: R's formula syntax auto-printed "`df$Revenue`" as the x-axis label; suppressed with `xlab = ""` since the group names already communicate the axis content. (2) Interaction plot: `col_pur` purple and `col_main` navy were adjacent cool hues, indistinguishable at small size or in grayscale; replaced with `col_green` (`#3A7D44`). The x-axis label and p-value annotation also overlapped at the bottom margin; label repositioned via `mtext()` and p-value moved to line 4.2, with bottom margin widened from 4 to 6.



Chapter 8: Key Findings, Limitations, and Future Work

8.1 Key Findings

Five research questions structured this analysis. The answers converge on a single behavioral logic: conversion is not random, and the signals that predict it are measurable within a session before any purchase decision is made.

PageValues is the dominant predictor. Most sessions never reach a page Google Analytics associates with revenue contribution. Buyers are overwhelmingly concentrated among those who do. No other variable in the dataset separates converters from non-converters as consistently, and that separation holds after controlling for browsing depth, visitor type, and temporal context in the logistic regression.

Depth and commercial exposure must co-occur. Time on product pages alone does not convert visitors. Reaching high-value pages alone does not either. Conversion concentrates in sessions where both conditions are simultaneously met. The k-means clustering makes this structural: the segment with the highest conversion rate is defined by exactly that combination, while the segment that matches on depth but not on commercial page exposure converts at a fraction of the rate. For practitioners, this means engagement metrics and page value metrics must be tracked together, not in isolation.

New visitors convert at higher per-session rates than returning visitors despite being a small minority of traffic. Returning visitors navigate to known pages and exit without broader exploration — familiarity produces targeted behavior, not purchasing behavior. New visitors who choose to arrive for the first time appear to do so with stronger purchase intent. Acquisition investment that focuses exclusively on volume, rather than the intent quality of incoming traffic, will systematically underperform.

Traffic source is the most variable categorical predictor of conversion. Some acquisition channels deliver visitors who convert at several times the site average. Others produce no purchases at all across thousands of sessions. This variation cannot be explained by session volume. It reflects genuine differences in the intent profiles of visitors arriving from different channels, and budget allocation that ignores this evidence misallocates resources.

Autumn converts most efficiently. The Black Friday and Cyber Monday window drives a conversion rate that no other season matches. Spring generates the most sessions but at a lower conversion rate. Volume and efficiency are decoupled across the calendar, and a promotional budget weighted by traffic volume rather than conversion efficiency will consistently underinvest in the highest-return period of the year.

8.2 Limitations

Four constraints bound these findings.

The dataset captures sessions, not customers. Repeat visit sequences, cart abandonment across sessions, and the relationship between visit frequency and eventual conversion are all questions this data cannot answer. Session-level patterns are real but incomplete as a picture of customer behavior.

January and April are absent without explanation. Two months of a twelve-month retail cycle are missing. Any retailer with significant traffic in those months cannot apply these findings directly without additional data.

Geographic and traffic type codes are anonymized. The conversion rate variation across channels is commercially significant but not actionable without the platform's original channel labels. A practitioner cannot reallocate budget toward high-converting traffic types when those types cannot be named.

Small subgroups produce imprecise estimates. The "Other" visitor category, several low-volume traffic types, and sparse regional cells for new visitors all carry implicit uncertainty that the descriptive analyses do not formally quantify. Findings for these subgroups should be treated as directional rather than definitive.

8.3 Future Work

Three directions follow from the findings.

The most immediate extension is customer-level longitudinal analysis. Linking sessions by user identifier would reveal how browsing behavior evolves across visits, how many sessions precede a first purchase, and whether first-session PageValues exposure predicts return visit conversion. The session-level patterns documented here would serve as a natural baseline for that comparison.

The HighIntent flag and EngagementScore feature identified in Chapter 5 are candidates for real-time intervention. If these behavioral signals can be detected within an active session rather than computed post-hoc, they could trigger targeted interventions at the moment of

highest purchase probability. Realizing that potential requires a properly withheld test set, a calibrated predictive model, and A/B testing infrastructure — none of which this study provides, but all of which the feature engineering work here directly supports.

The traffic type and regional patterns warrant follow-up with labeled data. If the platform's original channel identifiers can be recovered through UTM parameters, cost data, or channel-level attribution records, the conversion rate variation documented in Chapter 4 could be translated into channel-specific return on advertising spend estimates. That translation would convert a descriptive finding into a budget reallocation recommendation with direct commercial value.

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